Optimal geodesic cycles embedding of Möbius cubes *

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Abstract

For two vertices $X, Y \in V(G)$, a cycle is called a geodesic cycle with X and Y if a shortest path joining X and Y lies on the cycle. A graph G is called to be geodesic k-pancyclic if any two vertices X, Y on G have such geodesic cycle of length l that $2d_G(X,Y) + k \le l \le |V(G)|$. In this paper, we show that the n-dimensional Möbius cube MQ_n is geodesic 2-pancyclic for $n \ge 3$. This result is optimal in the sense that there is no geodesic 1-cycle with two adjacent vertices in MQ_n.

Keywords: geodesic pancyclic, Möbius cubes, panconnected, pancyclic, shortest path.

1 Introduction

A ring structure is often used as a interconnection architecture for local area network and as a control and data flow structure in distributed networks due to its good properties. To carry out a ring-structure algorithm on a specific multicomputer or a distributed system, the processes of the parallel algorithm need to be mapped to the nodes of the interconnection network in the system such that any two adjacent processes in the ring are mapped to two adjacent node of the network. For this purpose, it is desired that the targeted interconnection network posses a hamiltonian cycle, i.e., a cycle that passes every node of the network exactly once if the number of processes in the ring-structure parallel algorithm equals the number of nodes of the interconnection network. When the number of processes is less than the number of nodes of the network, the pancyclic property of the network with n nodes is desired, that is, there exists a cycle of length l in the network for each integer l with $4 \le l \le n$.

Throughout this paper, for the graph theoretic definitions and notations we follow [5]. Two vertices are adjacent when they are incident with a common edge. A path of length k from X to Y is a finite sequence of adjacent vertices written as $\langle X_1, X_1, X_2, \ldots, X_{k+1}\rangle$, where $X_1 = X$, $X_{k+1} = Y$, and all the vertices $X_1, X_1, X_2, \ldots, X_{k+1}$ are distinct except possibly $X_1 = X_{k+1}$. For convenience, we use the sequence $\langle X_1, \ldots, X_i,\ P(X_i, X_j), X_j, \ldots, X_{k+1}\rangle$ to denote the path $\langle X_1, X_2, \ldots, X_{k+1}\rangle$, where $P(X_i, X_j) = \langle X_i,\ X_{i+1},\ \ldots,\ X_j\rangle$ and the two vertices X_i and X_j are called the *end-vertices* of $P(X_i, X_j)$. We call that $P(X_i, X_j)$ is a *sub-path* of the path from X to Y. Let $l(P(X_i, X_j))$ denote the length of the path $P(X_i, X_j)$ that is the number of edges in $P(X_i, X_j)$. The *distance* between X and Y in G is denoted by $d_G(X, Y)$, which is the length of a shortest path between X and Y in G. A *cycle* C is a special path with at least three vertices such that the first vertex is the same as the last one. A cycle of length k is called a k-cycle.

The rest of this paper is organized as follows. In the next section, some fundamental definitions and notions are introduced. Section 3 shows that there exists a $2 \times 2^{n-1}$ mesh embedding in the n-dimensional Möbius cube. Section 4 proposes that two disjoint $4 \times 2^{n-3}$ meshes are embedded in n-dimensional 0-type Möbius cubes with dilation 1. The last section contains discussions and conclusions.

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2 Möbius cubes

The n-dimensional Möbius cube MQ_n, proposed first by Cull and Larson [2], consists of 2^n vertices and each vertex has a unique n-component binary vector for an address. Each vertex has n neighbors as follows. A vertex $X = x_1x_2 \ldots x_n$ connects to its ith neighbor, denoted by $N_i(X)$, for $2 \leq i \leq n$,

$$N_i(x) = x_nx_{n-1} \ldots x_{i+1}\overline{x}_ix_{i-1} \ldots x_1 \text{ if } x_{i+1} = 0$$

or

$$N_i(x) = x_nx_{n-1} \ldots x_{i+1}\overline{x}_i\overline{x}_{i-1} \ldots \overline{x}_1 \text{ if } x_{i+1} = 1.$$

For $i = 1$, since there is no bit on the left of x_1, $N_1(X)$ can be defined as the first neighbor of X can be denoted as $\overline{x}_1x_2 \ldots x_n$ or $\overline{x}_1\overline{x}_2 \ldots \overline{x}_n$. If we assume that the zeroth bit of every vertex of MQ_n is 0, we call the network a *0-type* n-dimensional Möbius cube, denoted by MQ_n^0; and if we assume that the zeroth bit of every vertex of MQ_n is 1, we call the network a *1-type* n-dimensional Möbius cube, denoted by MQ_n^1. Either MQ_n^0 or MQ_n^1 may be denoted by MQ_n. Therefore, MQ_n is an n-regular graph and can be recursively defined. MQ_n^0 and MQ_n^1 are both composed of a sub-Möbius cube MQ_{n-1}^0 and a sub-Möbius cube MQ_{n-1}^1. Each vertex $X = 0x_2x_3 \ldots x_{n-1}x_n \in V(MQ_{n-1}^0)$ connects to $1x_2x_3 \ldots x_{n-1}x_n \in V(MQ_{n-1}^1)$ in MQ_n^0; and to $1\overline{x}_2\overline{x}_3 \ldots \overline{x}_{n-1}\overline{x}_n \in V(MQ_{n-1}^1)$ in MQ_n^1. The examples of MQ_4^0 and MQ_4^1 are shown in Figure 1.

Let e_i be the n-dimensional $(0,1)$ vector with only its ith component equal to 1. Let E_i be the n-dimensional $(0,1)$ vector with ith through nth components equal to 1. Let $\{Z_2\}^n$ be the n-dimensional vector space over $\{0,1\}$ with addition and scalar multiplication mod 2. It is clear that both $\{e_i \mid 1 \leq i \leq n\}$ and $\{E_i \mid 1 \leq i \leq n\}$ are bases for this space. Hence $\{e_i, E_i \mid 1 \leq i \leq n\}$ forms a redundant basis for this vector space. Any vector X can be represented as a linear sum of these basis vectors:

$$X = \sum_{i=1}^{n}(\alpha_ie_i + \beta_iE_i), \qquad (1)$$

where $\alpha_i \in \{0,1\}$ and $\beta_i \in \{0,1\}$. Clearly, we can represent a vector X by the set of vectors e_i, E_i that have nonzero coefficients in the above sum. For any vertex X in MQ_n, the ith neighbor of X, $N_i(X)$, is formed by $X + e_i$ if $x_{i-1} = 0$, or $X + E_i$ if $x_{i-1} = 1$. It is clearly that every vertex X of MQ_n can be formulated by the above sum.

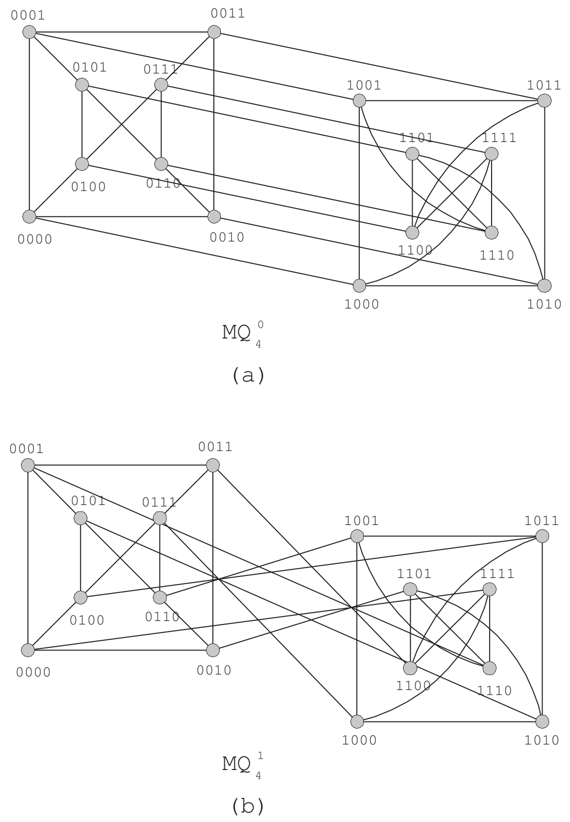

Figure 1: (a) A 0-type 4-dimensional Möbius cube. (b) A 1-type 4-dimensional Möbius cube.

Definition 1 *[2] A set S of e_i, E_i, where $1 \leq i \leq n$, is an expansion of X if and only if the equality in (1) is true, where $\alpha_i = 1$ if and only if $e_i \in S$ and $\beta_i = 1$ if and only if $E_i \in S$. Also, any $t \in S$ is called a term of this expansion of X.*

For a vector X, the weight of an expansion S of X is the cardinality of set S, denoted by $|S|$ and a minimal expansion of X is an expansion with least weight.

Definition 2 *[2] For a vector X, and term $t \in \{e_i, E_i \mid 1 \leq i \leq n\}$, t is a "bad" term with respect to X if and only if $(X, X+t)$ is not an edge of the Möbius cube.*

It is clearly, the only "bad" term with respect to X are e_i where $X_{i-1} = 1$ and E_i where $X_{i-1} = 0$.

Lemma 1 *[8] Assume that X and Y are two vertices of the n-dimensional 0-type Möbius cube MQ_n^0 with $n \geq 3$ and $Y = N_i(X)$ for some $2 \leq i \leq n$. Then $d_{MQ_n^0}(N_1(X), N_1(Y)) = 1$ if $3 \leq i \leq n$ and $d_{MQ_n^0}(N_1(X), N_1(Y)) = 2$ if $i = 2$.*

Lemma 2 *[8] Assume that X and Y are two vertices of the n-dimensional 1-type Möbius cube MQ_n^1 with $n \geq 3$ and $Y = N_i(X)$ for some $2 \leq i \leq n$. Then $d_{MQ_n^1}(N_1(X), N_1(Y)) = 1$ if $i = n$ and $d_{MQ_n^1}(N_1(X), N_1(Y)) = 2$ if $2 \leq i \leq n-1$.*

Cull and Larson [2] proposed an algorithm to generate a minimal expansion of a vector of X using only components x_i through x_n as $S(X, i)$ for $1 \leq i \leq n$.

Algorithm $S(X, i)$

Input: *A vector X and an integer i with $1 \leq i \leq n$.*

Output: *A minimal expansion of X using components x_i through x_n.*

begin

if $X = ()$ then return empty set.

if $X = (1)$ then return $\{E_i\}$.

if $X = (0X')$ then return $S(X', i+1)$.

if $X = (10X')$ then return $\{e_i\} \cup S(X', i+2)$.

if $X = (11X')$ then return $\{E_i\} \cup S(\overline{X'}, i+2)$.

end

Lemma 3 *[2] The "greedy" algorithm given in correctly produces a minimal expansion of X, by computing $S(X, 1)$.*

One can observe that E_{i+1} and e_{i+1} are not contained in $S(X, 1)$ if e_i or E_i is in $S(X, 1)$. Meanwhile, $S(X, 1)$ does not contain simultaneously E_i and e_i.

Lemma 4 *[2] Let X and Y be two vertices of MQ_n, and S be a minimal expansion of $X + Y$ produced by the "greedy" minimal expansion algorithm. Then $d_{MQ_n}(X, Y) = |S|$ or $|S| + 1$.*

According to an exact minimal routing algorithm given in [2], a shortest path $P_s(X, Y)$ between X and Y can be determined by examining each term of a "greedy" minimal expansion S in left to right order. Let $S = \{t_{j_1}, t_{j_2}, \ldots, t_{j_d} \mid t_{j_i} \in \{E_{j_i}, e_{j_i}\}$ where $1 \leq j_i \leq n$ and $j_i \leq j_{i+1}$ for $1 \leq i \leq d\}$. If $d_{MQ_n}(X, Y) = |S| = d$, $P_s(X, Y) = \langle X_1, X_2, \ldots, X_{d+1}\rangle$ where $X_1 = X$ and $X_{d+1} = Y$. Hence $X_{i+1} = X_i + t_{j_i}$ for $1 \leq i \leq d$. If $d_{MQ_n}(X, Y) = |S| + 1 = d + 1$, $P_s(X, Y) = \langle X_1, X_2, \ldots, X_{d+2}\rangle$ where $X_1 = X$ and $X_{d+2} = Y$. It is not difficult to verify that there exists a sub-path of $P_s(X, Y)$ denoted by $\langle X_i, X_{i+1}, X_{i+2}\rangle$ such that the minimal expansion of $X_i + X_{i+2}$ is $\{t_{j_i}\}$ and $t_{j_i} \in S$ is "bad" term with respect to X_i for some $1 \leq i \leq d$.

Lemma 5 *Let X and Y be two vertices of MQ_n, and S be a minimal expansion of $X + Y$ produced by the "greedy" minimal expansion algorithm. Then there exists a path $P_c(X, Y)$ of length $|S| + 2$ if $d_{MQ_n}(X, Y) = |S| + 1$.*

Proof. Let $S = \{t_{j_1}, t_{j_2}, \ldots, t_{j_d}\}$ where $t_{j_i} \in \{E_{j_i}, e_{j_i}\}$, $1 \leq j_i \leq n$ and $j_i \leq j_{i+1}$ for all i. Hence the shortest path $P_s(X, Y)$ is formed by $\langle X_1, X_2, \ldots, X_{d+2}\rangle$ where $X_1 = X$ and $X_{d+2} = Y$. Let the path $\langle X_i, X_{i+1}, X_{i+2}\rangle$ is a sub-path of $P_s(X, Y)$ such that the minimal expansion of $X_i + X_{i+2}$ is $\{t_{j_i}\}$ and t_{j_i} is "bad" with respect to X_i for some $1 \leq i \leq d$. Let $j_i = k$ and $X_i = a_1 a_2 \ldots a_n$. To create a path of length $d+2$ joining X and Y, we hope to find a path $\langle X_i, A, B, X_{i+2}\rangle$ such that A and B does not pass by original $P_s(X, Y)$ and replace the sub-path $\langle X_i, X_{i+1}, X_{i+2}\rangle$ with the path $\langle X_i, A, B, /X_{i+2}\rangle$. The proof is divided into two cases: (1) $1 \leq k \leq n-2$ and (2) $k = n-1$. Let S_i be the "greedy" minimal expansion of $X_i + X_{i+2}$.

Case 1: $1 \leq k \leq n-2$.

The proof of this case is partitioned into two parts.

Subcase 1.1: The term $t_k = e_k$ is "bad" with respect to X_i.

Obviously, $a_{k-1} = 1$, $X_i = a_1 a_2 \ldots a_{k-2} 1 a_k a_{k+1} \ldots a_n$, and $X_{i+2} = a_1 a_2 \ldots a_{k-2} 1 \overline{a}_k a_{k+1} \ldots a_n$. Note that $S_i = \{e_k\}$. One may replace e_k in S_i with the set $\{E_k, e_{k+1}, E_{k+2}\}$, and then routing from X_i to X_{i+2} with respect to the expansion $\{E_k, e_{k+1}, E_{k+2}\}$, the desired path is obtained. We only describe how to find this path if $a_k = 0$. Since the proof of the case of $a_k = 1$ is similar, details are omitted. Suppose that $a_k = 0$. We divided the proof into two subcases: **(1)** $a_{k+1} = 0$ and **(2)** $a_{k+1} = 1$.

Subcase 1.1.1: $a_{k+1} = 0$.

Hence $X_i = a_1 a_2 \ldots a_{k-2} 100 a_{k+2} \ldots a_n$ and $X_{i+2} = a_1 a_2 \ldots a_{k-2} 110 a_{k+2} \ldots a_n$. Let $A = X_i + e_{k+1} = a_1 a_2 \ldots a_{k-2} 101 a_{k+2} \ldots a_n$ and $B = A + E_{k+2} = a_1 a_2 \ldots a_{k-2} 101 \overline{a}_{k+2} \ldots \overline{a}_n$. This implies that $X_{i+2} = B + E_k = a_1 a_2 \ldots a_{k-2} 110 a_{k+2} \ldots a_n$. Hence $\langle X_i, A, B, X_{i+2}\rangle$ is a path in MQ_n.

Subcase 1.1.2: $a_{k+1} = 1$.

Hence $X_i = a_1 a_2 \ldots a_{k-2} 101 a_{k+2} \ldots a_n$ and $X_{i+2} = a_1 a_2 \ldots a_{k-2} 111 a_{k+2} \ldots a_n$. Let $B = X_{i+2} + E_k = a_1 a_2 \ldots a_{k-2} 100 \overline{a}_{k+2} \ldots \overline{a}_n$ and $A = B + e_{k+1} = a_1 a_2 \ldots a_{k-2} 101 \overline{a}_{k+2} \ldots \overline{a}_n$. Then implies $X_i = A + E_{k+2} = a_1 a_2 \ldots a_{k-2} 101 a_{k+2} \ldots a_n$. Hence $\langle X_i, A, B, X_{i+2}\rangle$ is a path in MQ_n.

Subcase 1.2 The term $t_k = E_k$ is "bad" with respect to X_i.

Obviously, $a_{k-1} = 0$, $X_i = a_1 a_2 \ldots a_{k-2} 0 a_k a_{k+1} \ldots a_n$, and $X_{i+2} =$

$a_1a_2\dots a_{k-2}0\overline{a}_k\overline{a}_{k+1}\dots\overline{a}_n$. Note that $S_i = \{E_k\}$. We replace E_k in S_i with the set $\{e_k, e_{k+1}, E_{k+2}\}$, and then routing from X_i to X_{i+2} with respect to the expansion $\{e_k, e_{k+1}, E_{k+2}\}$, the desired path is obtained. We only describe how to find this path if $a_k = 0$. Since the proof of the case of $a_k = 1$ is similar, details are omitted. Suppose that $a_k = 0$. We divided the proof into two subcases: **(1)** $a_{k+1} = 0$ and **(2)** $a_{k+1} = 1$.

Subcase 1.2.1: $a_{k+1} = 0$.

Hence $X_i = a_1a_2\dots a_{k-2}000a_{k+2}\dots a_n$ and $X_{i+2} = a_1a_2\dots a_{k-2}011\overline{a}_{k+2}\dots\overline{a}_n$. Let $B = X_{i+2}+E_{k+2} = a_1a_2\dots a_{k-2}011a_{k+2}\dots a_n$ and $A = B + e_k = a_1a_2\dots a_{k-2}001a_{k+2}\dots a_n$. Then implies $X_i = A + e_{k+1} = a_1a_2\dots a_{k-2}000a_{k+2}\dots a_n$. Hence $\langle X_i, A, B, X_{i+2}\rangle$ is a path in MQ_n.

Subcase 1.2.2: $a_{k+1} = 1$.

Hence $X_i = a_1a_2\dots a_{k-2}001a_{k+2}\dots a_n$ and $X_{i+2} = a_1a_2\dots a_{k-2}010\overline{a}_{k+2}\dots\overline{a}_n$. Let $A = X_i + E_{k+2} = a_1a_2\dots a_{k-2}001\overline{a}_{k+2}\dots\overline{a}_n$ and $B = A + e_{k+1} = a_1a_2\dots a_{k-2}000\overline{a}_{k+2}\dots\overline{a}_n$. This implies that $X_{i+2} = B + e_k/ = a_1a_2\dots a_{k-2}010\overline{a}_{k+2}\dots\overline{a}_n$. Hence $\langle X_i, A, B, X_{i+2}\rangle$ is a path in MQ_n.

Case 2: $k = n - 1$.

Suppose that $t_{n-1} = e_{n-1}$ is "bad" with respect to X_i. Let $S_i = \{e_{n-2}, e_{n-1}, e_{n-2}\}$ if $a_{n-3} = 0$ and $S_i = \{E_{n-2}, e_{n-1}, E_{n-2}\}$ if $a_{n-3} = 1$. Routing from X_i to X_{i+2} with respect to S_i, the desired path is obtained. Suppose that $t_{n-1} = E_{n-1}$ is "bad" with respect to X_i. Let $S_i = \{e_{n-2}, E_{n-1}, e_{n-2}\}$ if $a_{n-3} = 0$ and $S_i = \{E_{n-2}, E_{n-1}, E_{n-2}\}$ if $a_{n-3} = 1$. Routing from X_i to X_{i+2} with respect to S_i, the desired path is obtained. Since path construction schemes in this case is similar to Case 1, details are omitted.

Note that $P_s(X,Y) = \langle X_1, X_2, \dots, X_{i-1}, X_i, X_{i+1}, X_{i+2}, \dots, X_{d+2}\rangle$ is a shortest path joining X and Y where $X_1 = X$ and $X_{d+2} = Y$. Suppose that $A = X_j$ lies on the path $P_s(X,Y)$ for some $1 \le j \le d+2$. Since there is no triangle cycle in MQ_n, $A \ne X_i$ and $A \ne X_{i+1}$. Since $P_s(X,Y)$ is a shortest path between X and Y and (X_i, A) is an edge in MQ_n, $A \ne X_l$ for $i+2 \le l \le d+2$. If $j \le i-1$, there exists a path $P_c(X,Y)$ of $\langle X_1, X_2, \dots, X_j = A, B, X_{i+2}, X_{i+3}, \dots, X_{d+2}\rangle$ joining X and Y. Since $j \le i-1$, $l(P_c(X,Y)) \le d$. Then, $P_s(X,Y)$ is not a shortest path between X and Y that is a contradiction. Therefore, $P_s(X,Y)$ does not contain the vertex A. Similarly, one can verify that the vertex B does not lie on the path $P_s(X,Y)$. Hence the desired path of length $d+2$ is obtained. □

Lemma 6 *Let X and Y be two distinct vertices in MQ_n. Then $d_{MQ_n}(X,Y) = d_{MQ_n}(N_1(X), N_1(Y)) \pm k$ where $k = 0, 1$.*

Proof. Let $X = x_1x_2\dots x_n$ and $Y = y_1y_2\dots y_n$. Also let S be a minimal expansion of $X+Y$ produced by the "greedy" minimal expansion algorithm. It is observed that e_1 and E_1 doesn't be contained in S. Suppose that X and Y are in MQ_n^0. Hence $N_1(X) = \overline{x}_1x_2\dots x_n$ and $N_1(Y) = \overline{y}_1y_2\dots y_n$. Suppose that X and Y are in MQ_n^1. First neighbors of X and Y are $N_1(X) = \overline{x}_1\overline{x}_2\dots\overline{x}_n$ and $N_1(Y) = \overline{y}_1\overline{y}_2\dots\overline{y}_n$, respectively. One may observe that $X+Y = N_1(X)+N_1(Y)$. Consequently, S is a minimal expansion of $N_1(X)+N_1(Y)$. By Lemma 4, we have that $d_{MQ_n}(X,Y) = |S|$ or $d_{MQ_n}(X,Y) = |S|+1$, and $d_{MQ_n}(N_1(X), N_1(Y)) = |S|$ or $d_{MQ_n}(N_1(X), N_1(Y)) = |S|+1$. Therefore, $d_{MQ_n}(X,Y) = d_{MQ_n}(N_1(X), N_1(Y)) \pm k$ where $k = 0, 1$. □

It is without difficult to verify that Lemma 7-8 are true. Hence, we omitted detailed proofs of them.

Lemma 7 *Let X and Y be two vertices in the same sub-Möbius MQ_{n-1}^i of MQ_n with $i = 0, 1$. Then every shortest path $P_s(X,Y)$ joining X and Y in MQ_n satisfies that all vertices on $P_s(X,Y)$ belong to MQ_{n-1}^i.*

Lemma 8 *Let $X \in V(MQ_{n-1}^i)$ and $Y \in V(MQ_{n-1}^{1-i})$ be two vertices in MQ_n. Then there exists a shortest path $P_s(X,Y)$ of $\langle X, N_1(X), P_s(N_1(X), Y), Y\rangle$ or $\langle X, P_s(X, N_1(Y)), N_1(Y), Y\rangle$ where $P_s(X, N_1(Y))$ in MQ_{n-1}^i and $P_s(N_1(X), Y)$ in MQ_{n-1}^{1-i}.*

Lemma 9 *[7] If $n \ge 3$ then for any two distinct vertices X and Y in MQ_n, there exists a path of every length from $d_{MQ_n}(X,Y)+2$ to $2^n - 1$.*

3 MQ_n is geodesic 2-pancyclic

Definition 3 *Let G be a graph. For two vertices $X, Y \in V(G)$, a cycle is called a geodesic cycle with X and Y if a shortest path joining X and Y lies on the cycle. A geodesic l-cycle with X and Y in G, denoted by $gC^l(X,Y;G)$, is a geodesic cycle of length l.*

Definition 4 *Let G be a graph. For two vertices $X, Y \in V(G)$, they are called geodesic k-pancyclic on X and Y if for every integer l satisfying $2d_G(X,Y) + k \le l \le |V(G)|$, the geodesic cycle $gC^l(X,Y;G)$ exists.*

Definition 5 *The graph G is called geodesic k-pancyclic if any distinct two vertices on G are geodesic*

k-pancyclic on them. The geodesic-pancyclicity of G, denoted by $gpc(G)$, is defined as the minimum integer k such that G is geodesic k-pancyclic.

This section is dedicated to illustrating the geodesic pancyclic property of Möbius cubes. We first propose that MQ_3 is geodesic 2-pancyclic. Finally, we prove that $gpc(MQ_n) = 2$ for $n \geq 3$.

Lemma 10 *MQ_3 is geodesic 2-pancyclic.*

Proof. Since MQ_3^0 and MQ_3^1 are isomorphic, we only prove the case of MQ_3^0. Since MQ_3 is vertex-transitive. We assume that $X = 000$ and consider Y as the four cases: (1) $Y \in \{100, 010\}$, (2) $Y = 001$, (3) $Y = \{110, 111\}$, and (4) $Y \in \{101, 011\}$. By the symmetry of MQ_3, there is only one vertex is discussed for each case and related geodesic cycles are listed as Table 1.

Theorem 1 *MQ_n is geodesic 2-pancyclic for $n \geq 3$.*

Proof. The theorem is proved by induction on n. By Lemma 10, MQ_3 is geodesic 2-pancyclic. The theorem holds for $n = 3$. Assume that the theorem is true for every integer $3 \leq m < n$. We now consider $m = n$ as follows. Let X and Y be two vertices in MQ_n. By the relative position of X and Y, the proof is divided into two parts: **(1)** X and Y are in the same sub-Möbius MQ_{n-1}^i and **(2)** $X \in V(MQ_{n-1}^i)$ and $Y \in V(MQ_{n-1}^{1-i})$ for $i = 0, 1$.

Case 1: $X, Y \in V(MQ_{n-1}^i)$ for $i = 0, 1$.

Let $d_{MQ_n}(X, Y) = d$. Without loss of generality, we may assume that $X, Y \in V(MQ_{n-1}^0)$. By the induction hypothesis, we have the geodesic cycle $gC^l(X, Y; MQ_{n-1}^0)$ for all $2d_{MQ_{n-1}^0}(X, Y) + 2 \leq l \leq 2^{n-1}$. By Lemma 7, $d_{MQ_n}(X, Y) = d_{MQ_{n-1}^0}(X, Y) = d$. Therefore, the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $2d + 2 \leq l \leq 2^{n-1}$ follows.

We now construct the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $2^{n-1} + 1 \leq l \leq 2^{n-1}$. According to Lemma 6, $d_{MQ_{n-1}^1}(N_1(X), N_1(Y)) = d_{MQ_n}(N_1(X), N_1(Y)) = d$ or $d + 1$. By Lemma 9, there exists a path of $\langle N_1(Y)$,$P_c(N_1(Y), N_1(X))$, $N_1(X)\rangle$ in MQ_{n-1}^1 where $d_{MQ_n}(N_1(X), N_1(Y)) + 2 \leq l(P_c(N_1(Y), N_1(X)) \leq 2^{n-1} - 1$. By Lemma 5, there exists a path of length $d + 2$ joining $N_1(X)$ and $N_1(Y)$ if $d_{MQ_n}(N_1(X), N_1(Y)) = d + 1$. As a result, there exists a path of $\langle N_1(Y)$,$P_c(N_1(Y), N_1(X))$, $N_1(X)\rangle$ in MQ_{n-1}^1 where $d + 2 \leq l(P_c(N_1(Y), N_1(X)) \leq 2^{n-1} - 1$. Let cycle $C = \langle X$, $P_s(X, Y)$, Y, $N_1(Y)$, $P_c(N_1(Y), N_1(X))$, $N_1(X), X\rangle$. Then, $2d + 4 \leq l(C) \leq 2^{n-1} + d + 1$. We have that $2d + 4 \leq 2^{n-1} + 1$ for $n \geq 4$

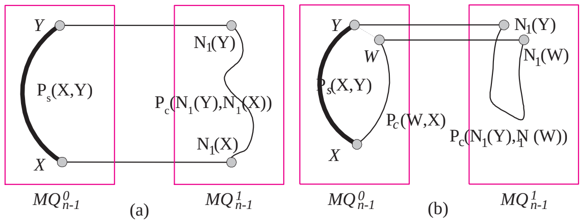

Figure 2: Two examples for case 1 of Theorem 1.

since $d \leq D(MQ_{n-1})$. Therefore, the geodesic cycle $gC^l(X, Y; MQ_n)$ exists where $2^{n-1} + 1 \leq l \leq 2^{n-1} + d + 1$. (See Figure 2 (a).)

Since $2^n - 1 > 2 \times D(MQ_n) + 2$ for $n \geq 3$, there exists the geodesic cycle $gC^{2^n-1}(X, Y; MQ_n)$ on any two distinct vertices X and Y in MQ_n for $n \geq 3$. Let $gC^{2^{n-1}-1}(X, Y; MQ_{n-1}^0)$=$\langle X$, $P_s(X, Y)$, Y, $P_c(Y, X)$, $X\rangle$ where $l(P_s(X, Y)) = d$. Let W and Y be two adjacent vertices on $P_c(Y, X)$. Hence $P_c(Y, X) = \langle Y, W, P_c(W, X), X\rangle$ where $W = N_j(Y)$ for some $2 \leq j \leq n$. By Lemma 6, $d_{MQ_n}(N_1(Y), N_1(W)) \leq 2$. By Lemma 9, there exists a path of $\langle N_1(Y)$, $P_c(N_1(Y), N_1(W))$, $N_1(W)\rangle$ in MQ_{n-1}^1 where $d_{MQ_n}(N_1(Y), N_1(W)) + 2 \leq l(P_c(N_1(Y), N_1(W))) \leq 2^{n-1} - 1$. By Lemma 5, there exists a path of length 3 joining $N_1(W)$ and $N_1(Y)$ if $d_{MQ_n}(N_1(W), N_1(Y)) = 2$. As a result, there exists a path of $\langle N_1(Y)$, $P_c(N_1(Y), N_1(W))$, $N_1(W)\rangle$ in MQ_{n-1}^1 where $3 \leq l(P_c(N_1(Y), N_1(W))) \leq 2^{n-1} - 1$. Let cycle $C = \langle X$, $P_s(X, Y)$, $Y, N_1(Y)$, $P_c(N_1(Y), N_1(W))$, $N_1(W), W, P_c(W, X), X\rangle$. Then $2^{n-1} + 3 \leq l(C) \leq 2^n - 1$. Since $d \geq 1$, we have the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $2^{n-1} + d + 2 \leq l \leq 2^n - 1$ with format C. (See Figure 2 (b).) Similarly, the geodesic cycle $gC^{2^n}(X, Y; MQ_n)$ may be obtained if the geodesic cycle $gC^{2^{n-1}}(X, Y; MQ_{n-1}^0)$ is used in the construction method. Hence, this case holds.

Case 2: $X \in V(MQ_{n-1}^i)$ and $Y \in V(MQ_{n-1}^{1-i})$ for $i = 0, 1$.

Without loss of generality, let $X \in V(MQ_{n-1}^0)$ and $Y \in V(MQ_{n-1}^1)$. According to relationship of X and Y, the proof of this case is divided into two parts: (1) $Y = N_1(X)$, i.e., X and Y are adjacent. (2) $Y \neq N_1(X)$, i.e., X and Y are not adjacent.

Subcase 2.1 $Y = N_1(X)$.

By Lemma 1-2, $N_n(Y)$ and $N_n(X)$ are adjacent. By Lemma 9, any path of $\langle N_n(Y)$, $P_c(N_n(Y), Y)$, $Y\rangle$ exists in MQ_{n-1}^1 where $1, 3 \leq l(P_c(N_n(Y), Y) \leq 2^{n-1} - 1$ and there exists a path of $\langle X$, $P_c(X, N_n(X))$, $N_n(X)\rangle$ in MQ_{n-1}^0 where $1, 3 \leq l(P_c(X, N_n(X)) \leq 2^{n-1} - 1$. Let cycle $C = \langle X, P_c(X, N_n(X)), N_n(X)$, $N_n(Y)$, $P_c(N_n(Y), Y)$, Y, $X\rangle$. Then $4, 6 \leq l(C) \leq$

Table 1: Summary of the geodesic cycles with $X = 000$ and Y in MQ_3^0.

Y	geodesic cyclic (even length)	geodesic cyclic (odd length)
100	$\langle 000, \underline{100}, 101, 001, 000\rangle$	$\langle 000, \underline{100}, 111, 011, 010, 000\rangle$
100	$\langle 000, \underline{100}, 111, 110, 101, 001, 000\rangle$	$\langle 000, \underline{100}, 101, 110, 111, 011, 001, 000\rangle$
100	$\langle 000, \underline{100}, 111, 011, 001, 101, 110, 010, 000\rangle$	
001	$\langle 000, \underline{001}, 011, 010, 000\rangle$	$\langle 000, \underline{001}, 011, 111, 100, 000\rangle$
001	$\langle 000, \underline{001}, 011, 111, 110, 010, 000\rangle$	$\langle 000, \underline{001}, 011, 111, 110, 101, 100, 000\rangle$
001	$\langle 000, \underline{001}, 011, 111, 100, 101, 110, 010, 000\rangle$	
110		$\langle 000, 010, \underline{110}, 111, 100, 000\rangle$
110	$\langle 000, 010, \underline{110}, 111, 011, 001, 000\rangle$	$\langle 000, 010, \underline{110}, 111, 100, 101, 001, 000\rangle$
110	$\langle 000, 010, \underline{110}, 101, 001, 011, 111, 100, 000\rangle$	
011	$\langle 000, 001, \underline{011}, 010, 000\rangle$	$\langle 000, 001, \underline{011}, 111, 100, 000\rangle$
011	$\langle 000, 001, \underline{011}, 111, 110, 010, 000\rangle$	$\langle 000, 001, \underline{011}, 111, 110, 101, 100, 000\rangle$
011	$\langle 000, 001, \underline{011}, 111, 100, 101, 110, 010, 000\rangle$	

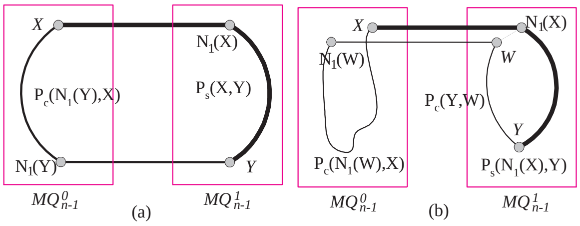

Figure 3: Two examples for subcase 2.2 of Theorem 1.

2^n. By Lemma 1-2, $d_{MQ_n}(Y, N_1(N_2(X))) = 2$, the geodesic cycle $gC^5(X, Y; MQ_n)$ can be found. Hence, we have the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $4 \leq l \leq 2^n$.

Subcase 2.2 $Y \neq N_1(X)$.

By Lemma 8, without loss of generality, we may assume there exists a shortest path $P_s(X, Y)$ with format of $\langle X, N_1(X), P_s(N_1(X), Y), Y\rangle$ where $P_s(N_1(X), Y)$ is a shortest path joining $N_1(X)$ and Y in MQ_{n-1}^1 (See Figure 3 (a)). Let $d_{MQ_n}(X, Y) = d$. Hence $d_{MQ_n}(N_1(X), Y) = d-1$. By Lemma 6, we have that $d_{MQ_n}(X, N_1(Y)) = d-1$ or d. By Lemma 9, there exists a path $P_c(N_1(Y), X)$ in MQ_{n-1}^0 where $d_{MQ_n}(X, N_1(Y)) + 2 \leq l(P_c(N_1(Y), X)) \leq 2^{n-1} - 1$. By Lemma 5, there exists a path of length $d+1$ joining X and $N_1(Y)$ if $d_{MQ_n}(X, N_1(Y)) = d$. Therefore, there exists a path $P_c(N_1(Y), X)$ in MQ_{n-1}^0 where $d+1 \leq l(P_c(N_1(Y), X)) \leq 2^{n-1} - 1$. Let cycle $C = \langle X, P_s(X, Y), Y, N_1(Y), P_c(N_1(Y), X), X\rangle$. Then, $2d+2 \leq l(C) \leq 2^{n-1} + d$. Consequently, there exists the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $2d+2 \leq l \leq 2^{n-1} + d$ with format C.

We now construct the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $2^{n-1} + d + 1 \leq l \leq 2^n$ (See Figure 3 (b)). By the induction hypothesis, $gC^l(N_1(X), Y; MQ_{n-1}^1)$ exists for all $2d \leq l \leq 2^{n-1}$. Let $gC^{2^{n-1}-1}(N_1(X), Y; MQ_{n-1}^1) = \langle N_1(X), P_s(N_1(X), Y), Y, P_c(Y, N_1(X)), N_1(X)\rangle$. Let W be the adjacent vertex of $N_1(X)$ on $P_c(Y, N_1(X))$. Hence $P_c(Y, N_1(X)) = \langle Y, P_c(Y, W), W, N_1(X)\rangle$. By Lemma 1-2, $d_{MQ_n}(X, N_1(W)) \leq 2$. By Lemma 9, there exists a path $P_c(N_1(W), X)$ in MQ_{n-1}^0 where $d_{MQ_n}(X, N_1(W)) + 2 \leq l(P_c(N_1(W), X)) \leq 2^{n-1} - 1$. By Lemma 5, there exists a path of length 3 if $d_{MQ_n}(X, N_1(W)) = 2$. Therefore, there exists a path $P_c(N_1(W), X)$ in MQ_{n-1}^0 where $3 \leq l(P_c) \leq 2^{n-1} - 1$. Let cycle $C = \langle X, N_1(X), P_s(N_1(X), Y), Y, P_c(Y, W), W, N_1(W), P_c(N_1(W), X), X\rangle$. Then the length of cycle C is $l(P_s(N_1(X), Y)) + l(P_c(Y, W)) + l(P_c(N_1(W), X)) + 2$. It is obvious that $2^{n-1} + 3 \leq l(C) \leq 2^n - 1$. Since $d \geq 2$, there exists the geodesic cycle $gC^l(X, Y; MQ_n)$ for all $2^{n-1} + d + 1 \leq l \leq 2^n - 1$. Similarly, the geodesic cycle $gC^{2^n}(X, Y; MQ_n)$ may be obtained if the geodesic cycle $gC^{2^{n-1}}(N_1(X), Y; MQ_{n-1}^1)$ is used in this construction method. Hence, this case holds.

It is well known that there is no triangle cycle in MQ_n. Therefore, there is no geodesic 1-cycle with two adjacent vertices in MQ_n. Then the following corollary holds.

Corollary 1 *$gpc(MQ_n) = 2$ for $n \geq 3$.*

A graph G is *edge-pancyclic* if for every edge $(X, Y) \in E(G)$ and every integer $4 \leq l \leq |V(G)|$, G contains a cycle of length l such that (X, Y) is in this cycle. Theorem 1 implies edge-pancyclic property of MQ_n for $n \geq 3$.

Corollary 2 *For $n \geq 3$, the n-dimensional Möbius cube is edge-pancyclic.*

4 Conclusions

In this paper, we demonstrate that for any two vertices X and Y in MQ_n for $n \geq 3$, there exists a geodesic l-cycle on them where $2d_{MQ_n}(X,Y)+2 \leq l \leq 2^n$. This result implies edge pancyclic property of MQ_n that has been proposed by Xu et al. [6].

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